

Lab 1: Packet analysis at application layer using Wireshark

SCSR1213 Network Communications

Universiti Teknologi Malaysia

Objective:

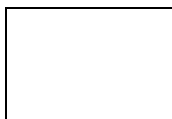
1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.
2. To introduce student with Wireshark software tool for packet analyzer.
3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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Metric No : A23CS0300

Section : 01



Mark

PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

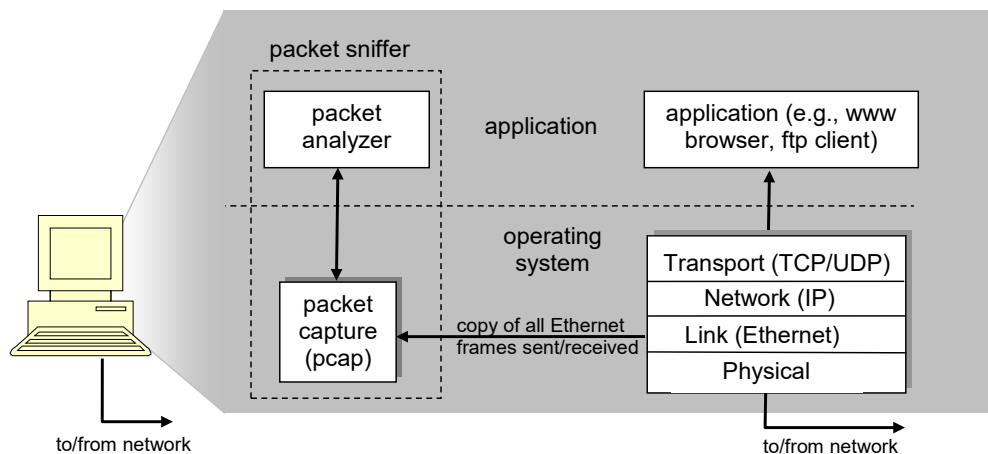


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

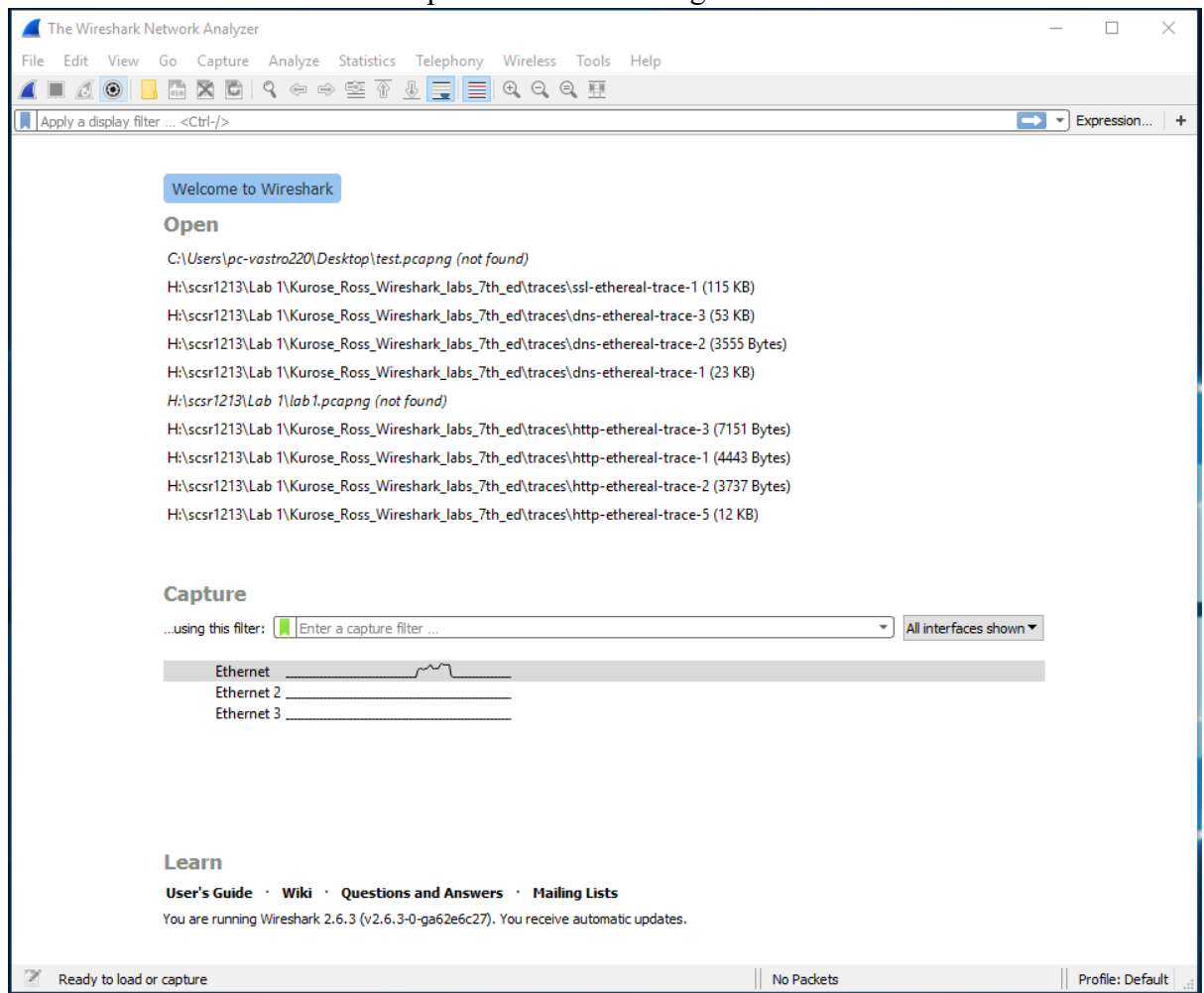


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

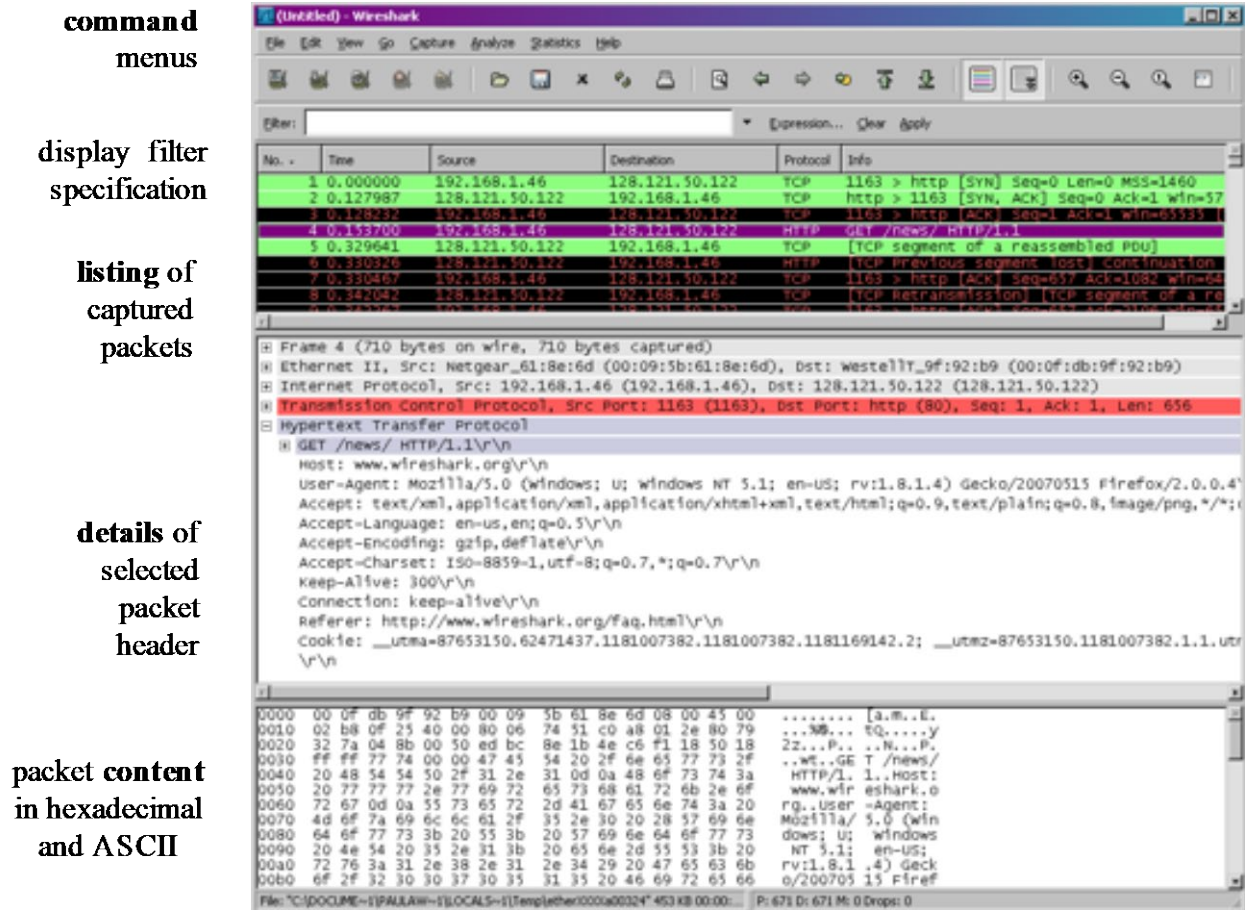


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

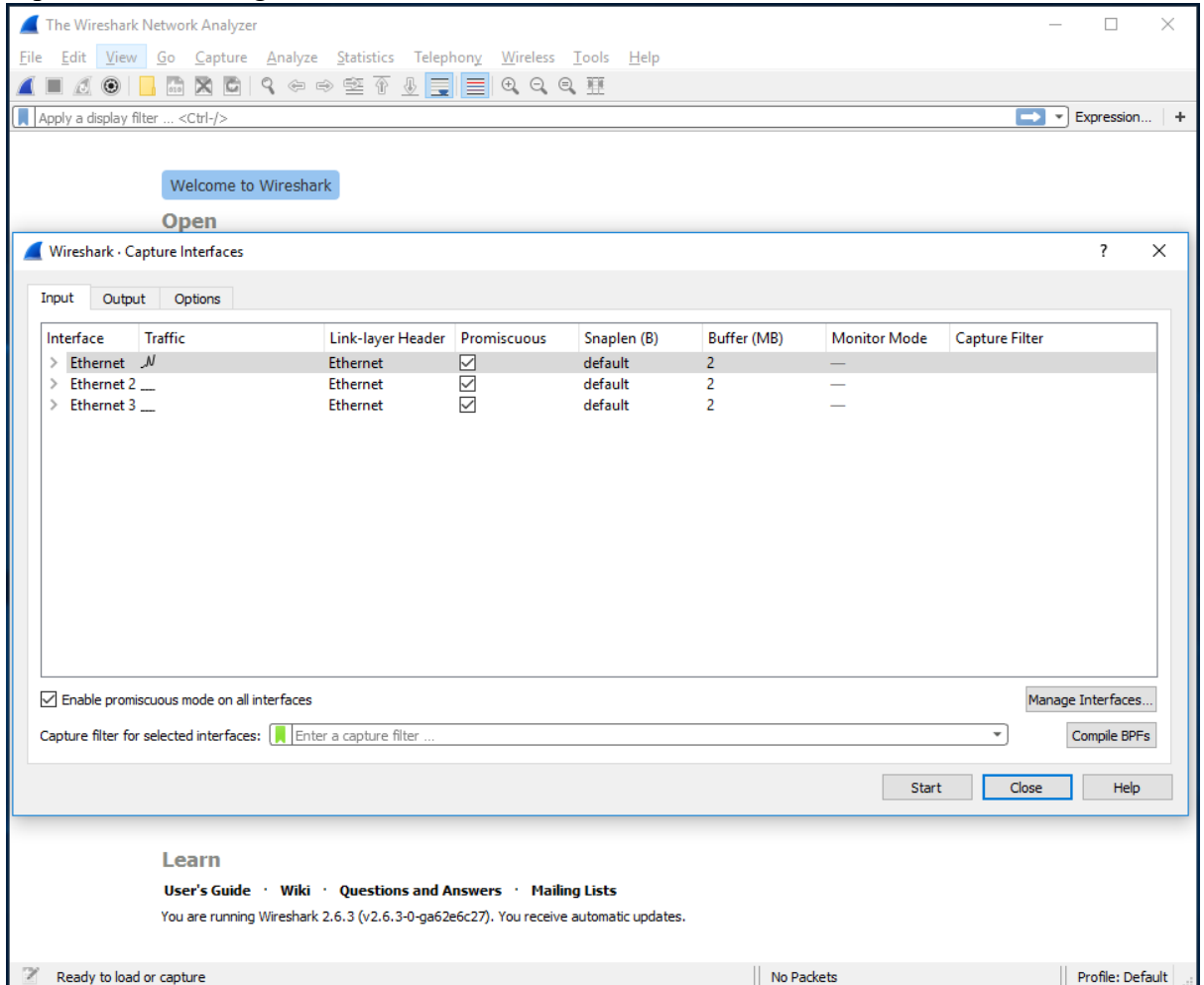


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

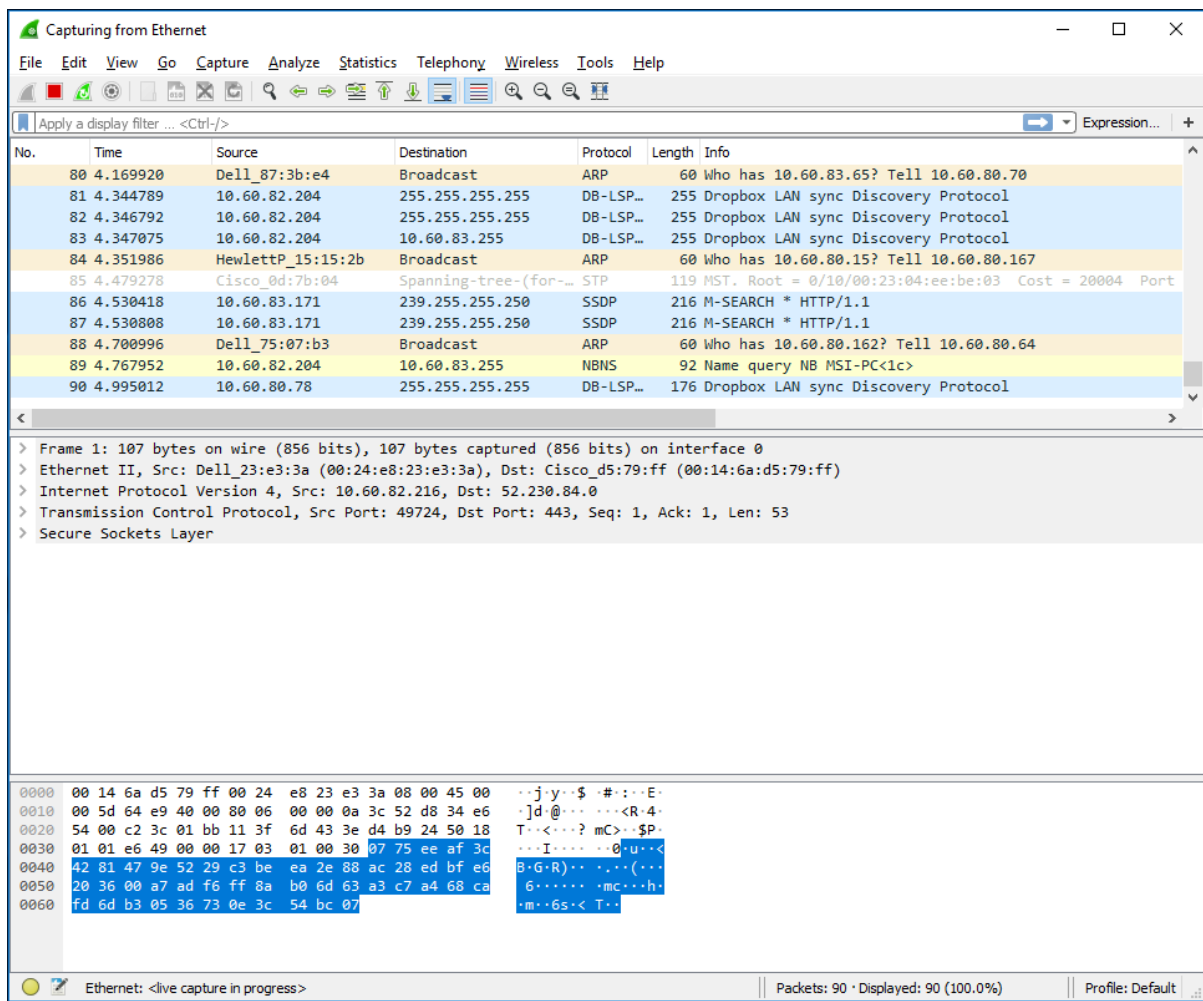


Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

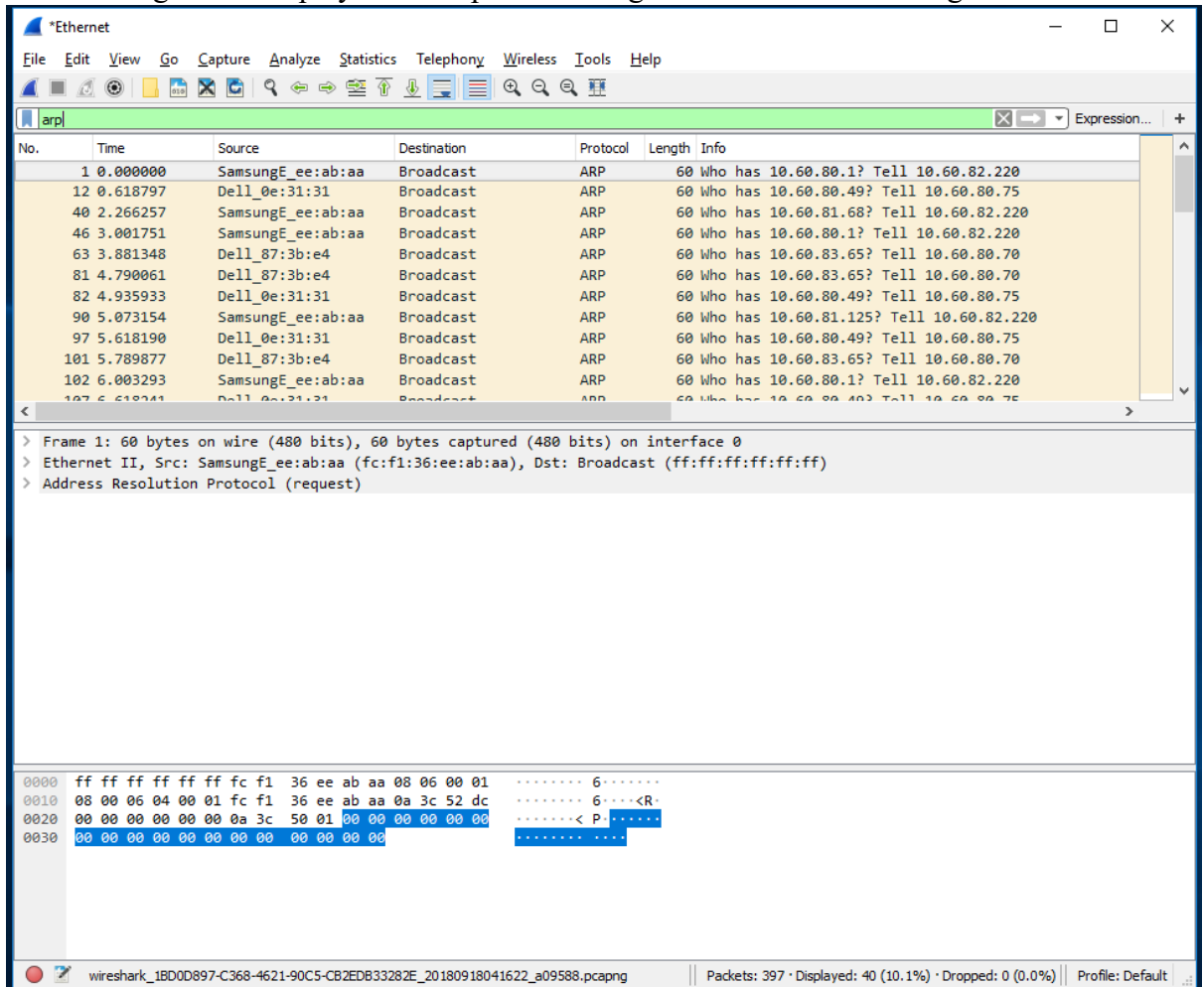


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

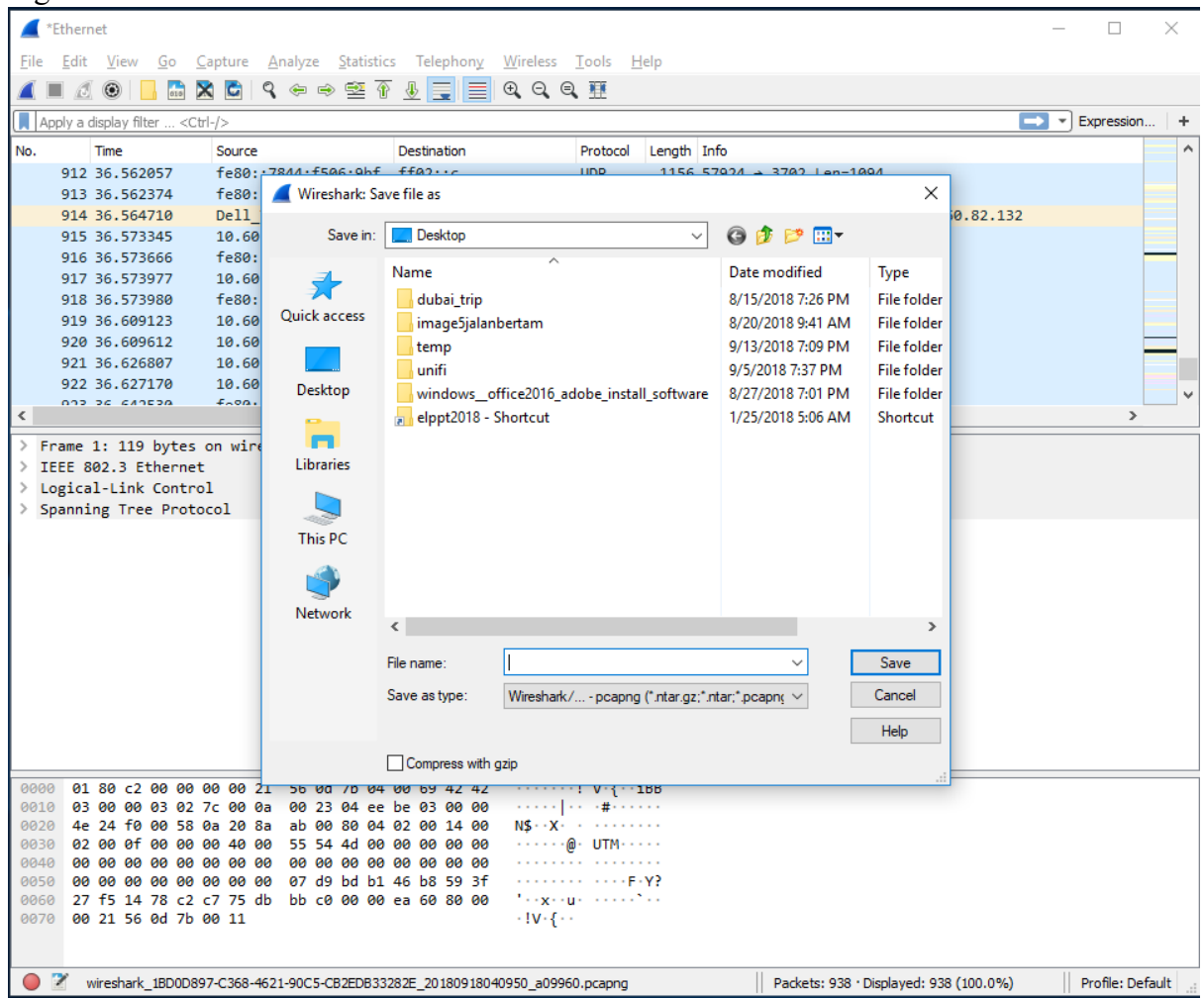


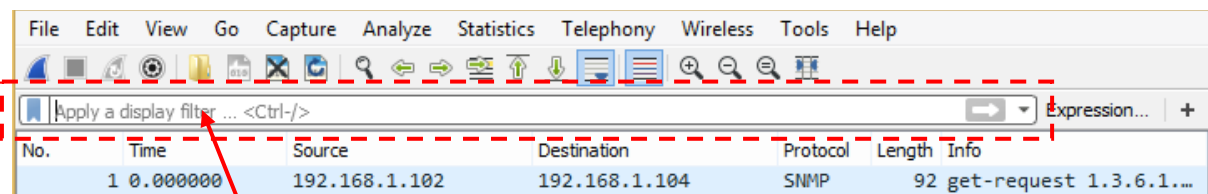
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter “**http**” (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



packet display filter

- By looking at the information in the HTTP GET and response messages, answer the following questions:

No.	Time	Source	Destination	Protocol	Length	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

1. What version of HTTP is the server running?
1.1
2. What is the IP address of the client computer?
192.168.1.102
3. What is the IP address of the gaia.cs.umass.edu server?
128.119.245.12

4. How many bytes of content are being returned to client browser?

73 bytes of contents are returned for status 200 OK and 1071 bytes of contents are returned for 404 Not Found

▼ Hypertext Transfer Protocol	▼ Hypertext Transfer Protocol
> HTTP/1.1 200 OK\r\n	> HTTP/1.1 404 Not Found\r\n
Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n	Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n
Server: Apache/2.0.40 (Red Hat Linux)\r\n	Server: Apache/2.0.40 (Red Hat Linux)\r\n
Last-Modified: Tue, 23 Sep 2003 05:29:00 GMT\r\n	Vary: accept-language\r\n
ETag: "1bfed-49-79d5bf00"\r\n	Accept-Ranges: bytes\r\n
Accept-Ranges: bytes\r\n	> Content-Length: 1071\r\n
> Content-Length: 73\r\n	Keep-Alive: timeout=10, max=99\r\n
Keep-Alive: timeout=10, max=100\r\n	Connection: Keep-Alive\r\n
Connection: Keep-Alive\r\n	Content-Type: text/html; charset=ISO-8859-1\r\n
Content-Type: text/html; charset=ISO-8859-1\r\n	\r\n
\r\n	

5. What is the status code returned from the server to client browser?

200 (OK) and 404 (Not Found)

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

No, there is no “IF-MODIFIED-SINCE” line in the HTTP GET.

No.	Source	Destination	Protocol	Length	Info
1	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-2.html HTTP/1.1
2	128.119.245.12	192.168.1.102	HTTP	739	HTTP/1.1 200 OK (text/html)
3	192.168.1.102	128.119.245.12	HTTP	668	GET /ethereal-labs/lab2-2.html HTTP/1.1
4	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified

> Frame 8: 555 bytes on wire (4440 bits), 555 bytes captured (4440 bits) on interface unknown, id 0	0000 00 06 25 da af 73 00 08 74 4f 36 23 08 00 45 00	..%...s... t068--E-
> Ethernet II, Src: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: LinksysGroup_da:af:73 (00:06:25:da:af:73)	0010 02 1d 02 5a 40 00 00 06 00 00 c0 a8 01 66 80 77	...Z@... ..f-w
> Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.119.245.12	0020 f5 0c 10 97 00 50 fa 88 01 31 81 6a b3 81 50 18	P... 1.j..P-
> Transmission Control Protocol, Src Port: 4247, Dst Port: 80, Seq: 1, Ack: 1, Len: 501	0030 fa f0 39 a2 00 00 47 45 54 20 2f 65 74 68 65 72	..9...GE T /ether
> Hypertext Transfer Protocol	0040 65 61 6c 2d 6c 61 62 73 2f 6c 61 62 32 2d 32 2e	...eal-labs /lab2-2.
	0050 68 74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a 48	html HTTP/1.1. H
	0060 6f 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d 61	ost: gai a.cs.uma
	0070 73 73 2e 65 64 75 0d 0a 55 73 65 72 2d 41 67 65	ss.edu... User-Age
	0080 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20	nt: Mozi lla/5.0
	0090 28 57 69 6e 64 6f 77 73 3b 20 55 3b 20 57 69 6e	(Windows ; U; Win
	00a0 64 6f 77 73 20 46 54 20 35 2e 31 3b 20 65 6e 2d	dows HT 5.1; en
	00b0 55 53 3b 20 72 76 3a 31 2e 30 2e 32 29 20 47 65	US; rv:1 .0.2) Ge
	00c0 63 6b 6f 2f 32 30 30 32 31 31 32 30 20 4e 65 74	cko/2002 1120 Net
	00d0 73 63 61 70 65 2f 37 2e 30 31 0d 0a 41 63 63 65	scape/7. 01..Acce
	00e0 70 74 3a 20 74 65 78 74 2f 78 6d 6c 2c 61 70 70	pt: text /xml,app
	00f0 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c 2c 61 70 70	lication /xml,app
	0100 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c 2c 61 70 70	lication /html;ex
	0110 6d 6c 2c 74 65 78 74 2f 68 74 6d 6c 3b 71 3d 30	ml;text/ html;q=0
	0120 2e 39 2c 74 65 78 74 2f 70 6c 61 69 6e 3b 71 3d	.9;text/ plain;q=
	0130 30 2e 38 2c 76 69 64 65 6f 2f 78 2d 6d 6e 67 2c	0.8,vide o/x-mng
	0140 69 6d 61 67 65 2f 70 6e 67 2c 69 6d 61 67 65 2f	image/pn g,image/
	0150 6a 70 65 67 2c 69 6d 61 67 65 2f 67 69 66 3b 71	jpeg,ima ge/gif;q
	0160 3d 30 2e 32 2c 74 65 78 74 2f 63 73 73 2c 2a 2f	=0.2,tex t/css;*/

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

Yes it did. It sent 200 OK message, which tells the client nothing is wrong. It also sent 371 bytes (content-length).

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

Yes, the If-Modified-Since is present because the site had been visited before. The information is If-Modified-Since : Tue, 23 Sep 2003 05:35:00 GMT\r\n

Connection: keep-alive\r\n

If-Modified-Since: Tue, 23 Sep 2003 05:35:00 GMT\r\n

If-None-Match: "1bfef-173-8f4ae900"\r\n

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

The status code is 304 Not Modified. The server did not explicitly return the contents of the file because it was not modified. Hence, no bytes were sent (no content-length specified).

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?

Client browser had sent 3 HTTP GET request messages

http						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-4.html HTTP/1.1
2	0.000000	128.119.245.12	192.168.1.102	HTTP	1057	HTTP/1.1 200 OK (text/html)
3	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
4	7.308803	192.168.1.102	134.241.6.82	HTTP	609	GET /~kurose/cover.jpg HTTP/1.1
5	7.333054	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (GIF89a)
6	7.589877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document follows (JPEG JFIF image)

2. To which Internet addresses were these GET requests sent?

1st GET IP address : 128.119.245.12

2nd GET IP address : 165.193.123.218

3rd GET IP address : 134.241.6.82

3. any bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?

Yes. It sent 3357 bytes of content length.

4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?

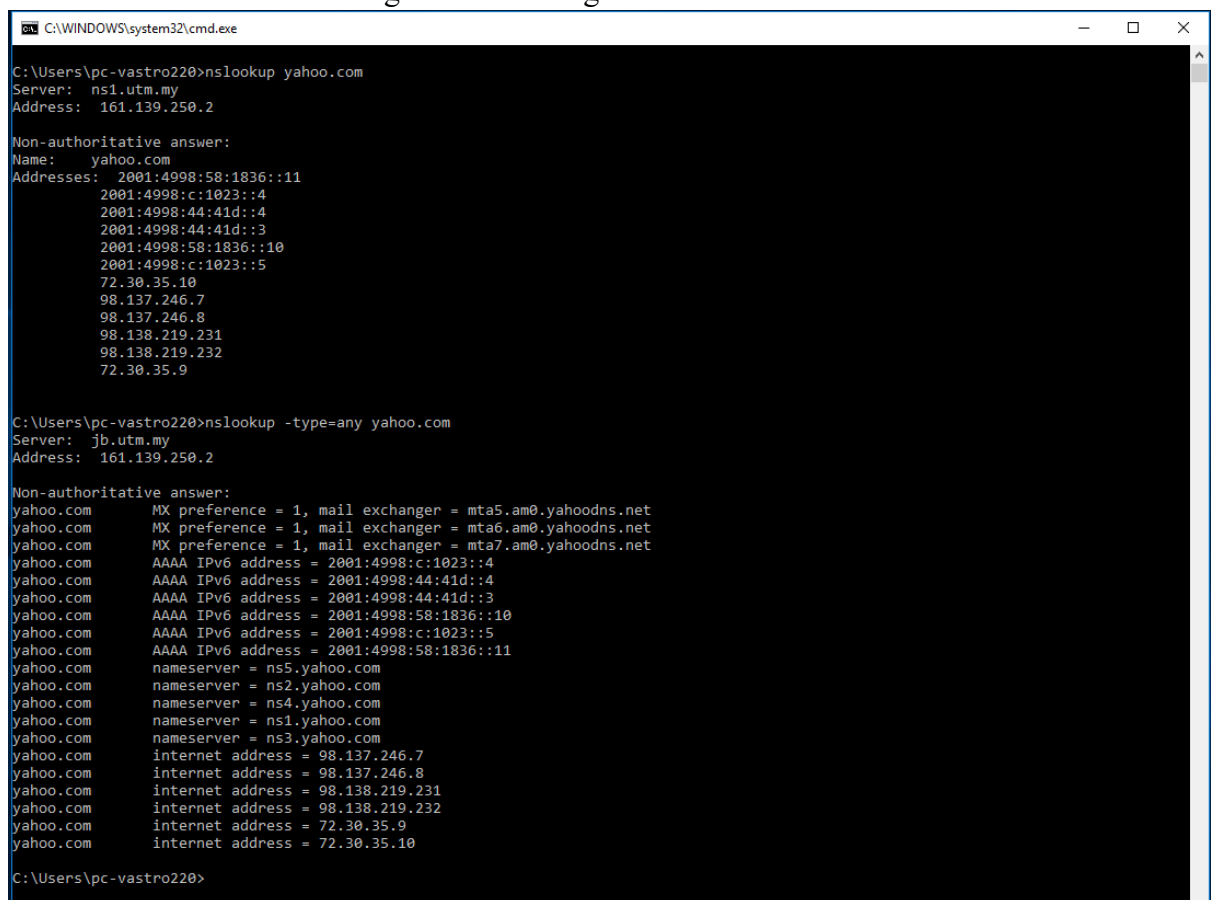
15642 bytes of content length.

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server: ns1.utm.my
Address: 161.139.250.2

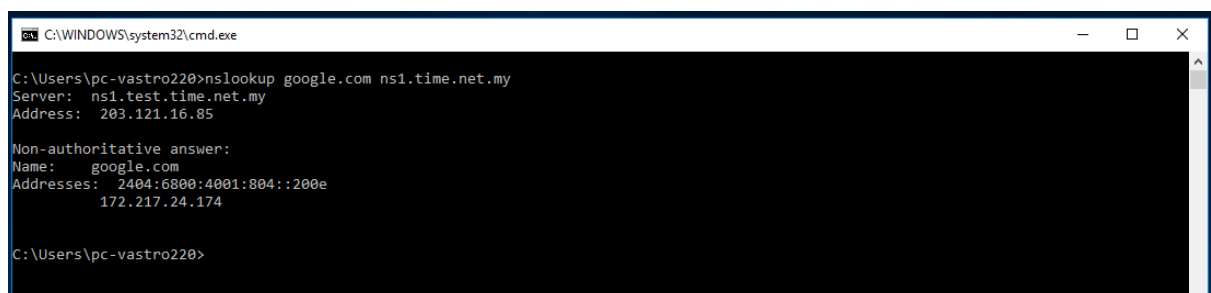
Non-authoritative answer:
Name: yahoo.com
Addresses: 2001:4998:58:1836::11
           2001:4998:c:1023::4
           2001:4998:44:41d::4
           2001:4998:44:41d::3
           2001:4998:58:1836::10
           2001:4998:c:1023::5
           72.30.35.10
           98.137.246.7
           98.137.246.8
           98.138.219.231
           98.138.219.232
           72.30.35.9

C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server: jlb.utm.my
Address: 161.139.250.2

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com      nameserver = ns5.yahoo.com
yahoo.com      nameserver = ns2.yahoo.com
yahoo.com      nameserver = ns4.yahoo.com
yahoo.com      nameserver = ns1.yahoo.com
yahoo.com      nameserver = ns3.yahoo.com
yahoo.com      internet address = 98.137.246.7
yahoo.com      internet address = 98.137.246.8
yahoo.com      internet address = 98.138.219.231
yahoo.com      internet address = 98.138.219.232
yahoo.com      internet address = 72.30.35.9
yahoo.com      internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server: ns1.test.time.net.my
Address: 203.121.16.85

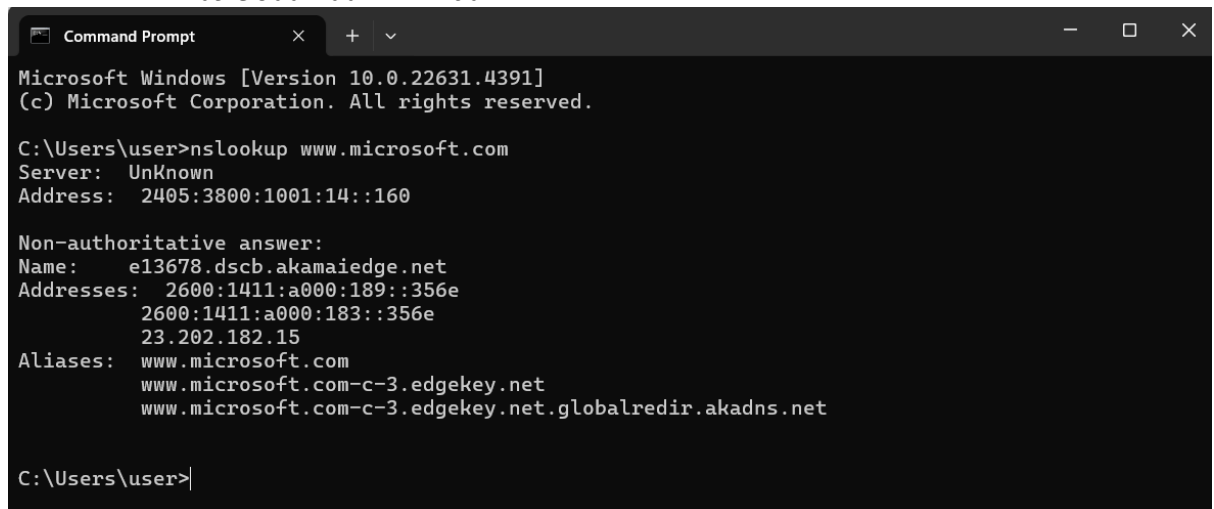
Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4001:804::200e
           172.217.24.174

C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

IP address : 2405:3800:1001:14::160



```
Command Prompt
Microsoft Windows [Version 10.0.22631.4391]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>nslookup www.microsoft.com
Server: UnKnown
Address: 2405:3800:1001:14::160

Non-authoritative answer:
Name: e13678.dscb.akamaiedge.net
Addresses: 2600:1411:a000:189::356e
           2600:1411:a000:183::356e
           23.202.182.15
Aliases: www.microsoft.com
          www.microsoft.com-c-3.edgekey.net
          www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net

C:\Users\user>
```

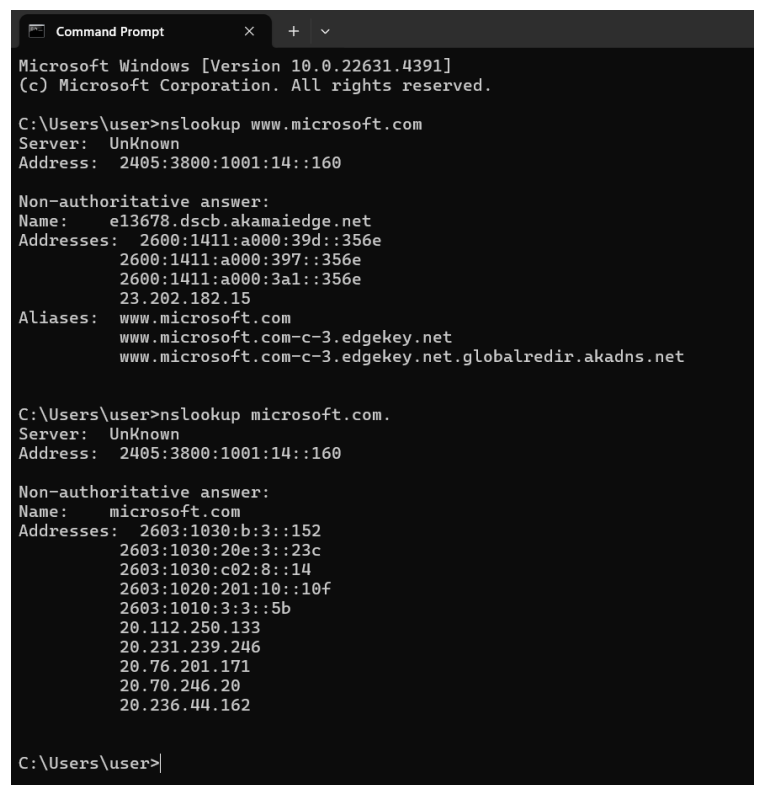
2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.

The non-authoritative DNS servers for domain microsoft.com is UnKnown with IP address 2405:3800:1001:14::160

Non-authoritative answer:

Name: microsoft.com

Addresses: 2603:1030:b:3::152
2603:1030:20e:3::23c
2603:1030:c02:8::14
2603:1020:201:10::10f
2603:1010:3:3::5b
20.112.250.133
20.231.239.246
20.76.201.171
20.70.246.20
20.236.44.162



```
Command Prompt
Microsoft Windows [Version 10.0.22631.4391]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>nslookup www.microsoft.com
Server: UnKnown
Address: 2405:3800:1001:14::160

Non-authoritative answer:
Name: e13678.dscb.akamaiedge.net
Addresses: 2600:1411:a000:39d::356e
           2600:1411:a000:397::356e
           2600:1411:a000:3a1::356e
           23.202.182.15
Aliases: www.microsoft.com
          www.microsoft.com-c-3.edgekey.net
          www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net

C:\Users\user>nslookup microsoft.com.
Server: UnKnown
Address: 2405:3800:1001:14::160

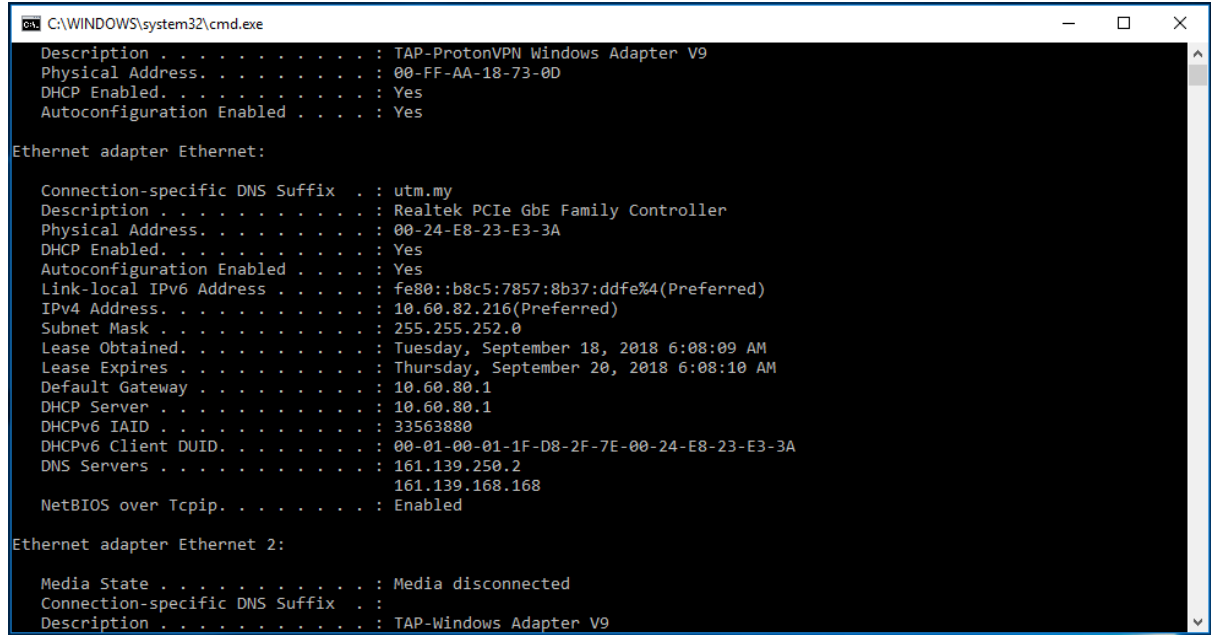
Non-authoritative answer:
Name: microsoft.com
Addresses: 2603:1030:b:3::152
           2603:1030:20e:3::23c
           2603:1030:c02:8::14
           2603:1020:201:10::10f
           2603:1010:3:3::5b
           20.112.250.133
           20.231.239.246
           20.76.201.171
           20.70.246.20
           20.236.44.162

C:\Users\user>
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all



```
C:\WINDOWS\system32\cmd.exe
Description . . . . . : TAP-ProtonVPN Windows Adapter V9
Physical Address. . . . . : 00-FF-AA-18-73-0D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

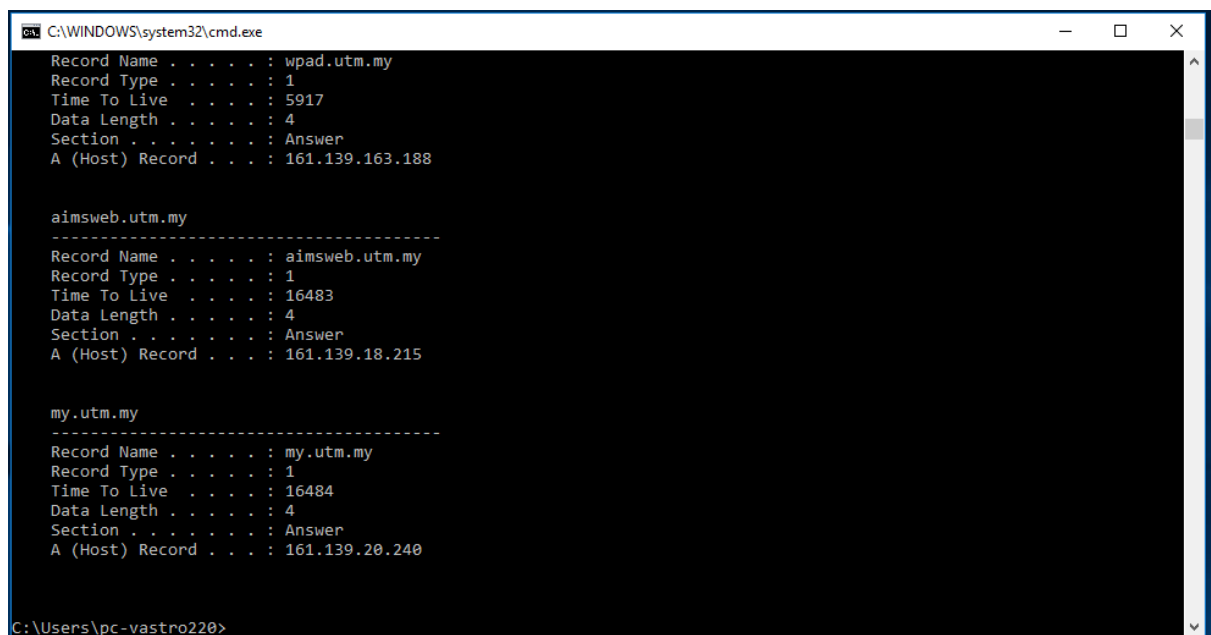
    Connection-specific DNS Suffix  . : utm.my
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : 00-24-E8-23-E3-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
    IPv4 Address. . . . . : 10.60.82.216(Preferred)
    Subnet Mask . . . . . : 255.255.252.0
    Lease Obtained. . . . . : Tuesday, September 18, 2018 6:08:09 AM
    Lease Expires . . . . . : Thursday, September 20, 2018 6:08:10 AM
    Default Gateway . . . . . : 10.60.80.1
    DHCP Server . . . . . : 10.60.80.1
    DHCPv6 IAID . . . . . : 33563880
    DHCPv6 Client DUID. . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
    DNS Servers . . . . . : 161.139.250.2
                           161.139.168.168
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
    Description . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds.
Command: ipconfig /displaydns



```
C:\WINDOWS\system32\cmd.exe
Record Name . . . . . : wpad.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 5917
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.163.188

-----
Record Name . . . . . : aimswb.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16483
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.18.215

-----
Record Name . . . . . : my.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16484
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.20.240

C:\Users\pc-vastro220>
```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns

```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\pc-vastro220>
```

Figure C.5: ipconfig /flushdns result

3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1. Answer the following questions.
- 1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

UDP (Query)

No.	Time	Source	Destination	Protocol	Length	Info
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51

Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits) on interface 0

Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)

Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 58

Identification: 0x229e (8862)

> 000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 128

Protocol: UDP (17)

Header Checksum: 0xd281 [validation disabled]

[Header checksum status: Unverified]

Source Address: 128.238.38.160

Destination Address: 128.238.29.23

[Stream index: 1]

UDP (Response)

No.	Time	Source	Destination	Protocol	Length	Info
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51

Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits) on interface 0

Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)

Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0xd595 (DSCP: CS0, ECN: Not-ECT)

Total Length: 90

Identification: 0xd595 (54677)

> 000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 126

Protocol: UDP (17)

Header Checksum: 0x216a [validation disabled]

[Header checksum status: Unverified]

Source Address: 128.238.29.23

Destination Address: 128.238.38.160

[Stream index: 1]

Use Datastream Protocol. Src Port: 53. Dst Port: 3163

- What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

destination port for the DNS query message : 53

```
> Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
✓ User Datagram Protocol, Src Port: 3163, Dst Port: 53
  Source Port: 3163
  Destination Port: 53
  Length: 38
  Checksum: 0x8acb [unverified]
  [Checksum Status: Unverified]
  [Stream index: 1]
  [Stream Packet Number: 1]
  > [Timestamps]
  UDP payload (30 bytes)
> Domain Name System (query)
```

the source port of DNS response message : 53

```
> Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
> Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
> Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
✓ User Datagram Protocol, Src Port: 53, Dst Port: 3163
  Source Port: 53
  Destination Port: 3163
  Length: 70
  Checksum: 0xb0ba [unverified]
  [Checksum Status: Unverified]
  [Stream index: 1]
  [Stream Packet Number: 2]
  > [Timestamps]
  UDP payload (62 bytes)
> Domain Name System (response)
```

- To what IP address is the DNS query message sent? Add screenshots in your answer.
IP address : 128.238.29.23

No.	Time	Source	Destination	Protocol	Length	Info
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51

- Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

DNS query is Type A, and the query message does not contain any answer

```
> Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
> User Datagram Protocol, Src Port: 3163, Dst Port: 53
✓ Domain Name System (query)
  Transaction ID: 0x006e
  > Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  ✓ Queries
    > www.ietf.org: type A, class IN
    [Response In: 9]
```

5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

DNS response message contains 2 answers which is

1st answer : www.ietf.org: type A, class IN, addr 132.151.6.75

2nd answer : www.ietf.org: type A, class IN, addr 65.246.255.51

```
> Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
> Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
> Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
> User Datagram Protocol, Src Port: 53, Dst Port: 3163
✓ Domain Name System (response)
  Transaction ID: 0x006e
  > Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 2
  Authority RRs: 0
  Additional RRs: 0
  ✓ Queries
    > www.ietf.org: type A, class IN
  ✓ Answers
    ✓ www.ietf.org: type A, class IN, addr 132.151.6.75
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 1678 (27 minutes, 58 seconds)
      Data length: 4
      Address: 132.151.6.75
    ✓ www.ietf.org: type A, class IN, addr 65.246.255.51
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 1678 (27 minutes, 58 seconds)
      Data length: 4
      Address: 65.246.255.51
  [Request In: 8]
  [Time: 0.000844000 seconds]
```

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.

Yes. The destination of IP address is 132.151.6.75 which is correspond to the address provided by the DNS server for www.ietf.org.

No.	Time	Source	Destination	Protocol	Length	Info
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1
14	3.111678	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=0
15	3.120640	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
16	3.120693	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1381 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
17	3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=2761 Win=64860 Len=0
18	3.148016	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=2761 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
19	3.148069	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=4141 Win=64860 Len=0
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055	HTTP/1.1 200 OK (text/html)
21	3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=5143 Win=63859 Len=0
22	3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK] Seq=376 Ack=5143 Win=63859 Len=0
23	3.174716	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=5143 Ack=377 Win=6432 Len=0
24	3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
25	3.179283	128.238.38.160	132.151.6.75	TCP	62	3371 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
26	3.191649	132.151.6.75	128.238.38.160	TCP	62	80 → 3370 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
27	3.191726	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
28	3.191998	128.238.38.160	132.151.6.75	HTTP	320	GET /images/ietflogo2e.gif HTTP/1.1
29	3.192665	132.151.6.75	128.238.38.160	TCP	62	80 → 3371 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
30	3.192695	128.238.38.160	132.151.6.75	TCP	54	3371 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
31	3.192869	128.238.38.160	132.151.6.75	HTTP	314	GET /images/blue.eif HTTP/1.1

```
> Frame 10: 62 bytes on wire (496 bits), 62 bytes captured (496 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 132.151.6.75
> Transmission Control Protocol, Src Port: 3369, Dst Port: 80, Seq: 0, Len: 0
```

```
0000 00 00 0c 07 ac 00 00 09 6b 10 60 99 00 00 45 00 .....k.....E
0010 00 30 22 9f 40 00 00 06 a5 b8 80 ee 26 a0 84 97 0* @... ..&...
0020 06 4b 0d 29 00 50 f2 bd 56 0c 00 00 00 70 02 .K.)P..V.....
0030 fa f0 ff 7a 00 00 02 04 05 b4 01 01 04 02 .....<.....<
```

7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

No, my host does not issue new DNS queries because DNS query will only appear once in the entire tracing. There is no new DNS queries issue by host between line number 10 and number 27 before retrieving image in line number 28 as shown.

No.	Time	Source	Destination	Protocol	Length	Info
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1
14	3.111678	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=0
15	3.120640	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
16	3.128093	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1381 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
17	3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=2761 Win=64860 Len=0
18	3.148016	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=2761 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
19	3.148069	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=4141 Win=64860 Len=0
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055	HTTP/1.1 200 OK (text/html)
21	3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=5143 Win=63859 Len=0
22	3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK] Seq=376 Ack=5143 Win=63859 Len=0
23	3.174716	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=5143 Ack=377 Win=6432 Len=0
24	3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
25	3.179283	128.238.38.160	132.151.6.75	TCP	62	3371 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
26	3.191649	132.151.6.75	128.238.38.160	TCP	62	80 → 3370 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
27	3.191726	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
28	3.191998	128.238.38.160	132.151.6.75	HTTP	320	GET /images/ietflogo2e.gif HTTP/1.1

- Open packet trace file dns-trace-2 for nslookup.
 - We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
 - Answer the following questions.
8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.
destination port for the DNS query message : 53

No.	Time	Source	Destination	Protocol	Length	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.231
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.ed
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.ed

> Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22
> User Datagram Protocol, Src Port: 3742, Dst Port: 53
Source Port: 3742
Destination Port: 53
Length: 37
Checksum: 0x5890 [unverified]
[Checksum Status: Unverified]
[Stream index: 3]
[Stream Packet Number: 1]
> [Timestamps]
UDP payload (29 bytes)
> Domain Name System (query)

the source port of DNS response message : 53

dns						
No.	Time	Source	Destination	Protocol	Length	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.a
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18

> Frame 20: 196 bytes on wire (1568 bits), 196 bytes captured (1568 bits)

> Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)

> Internet Protocol Version 4, Src: 128.238.29.22, Dst: 128.238.38.160

> User Datagram Protocol, Src Port: 53, Dst Port: 3742

Source Port: 53

Destination Port: 3742

Length: 162

Checksum: 0xa318 [unverified]

[Checksum Status: Unverified]

[Stream index: 3]

[Stream Packet Number: 2]

> [Timestamps]

UDP payload (154 bytes)

> Domain Name System (response)

9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.
IP address that sent by DNS query message is 128.238.29.22. While my default local DNS server IP address is 192.168.0.1.

dns						
No.	Time	Source	Destination	Protocol	Length	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRAWB.mit.edu NS W

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : 
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : 8C-B8-7E-C4-CD-18
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2001:3::9f7:79ae:34ba:dc78(Preferred)
Temporary IPv6 Address. . . . . : 2001:3::c51f:4a83:8675:9855(Preferred)
Link-local IPv6 Address . . . . . : fe80::e000:c99b:7360:4b42%5(Preferred)
IPv4 Address. . . . . : 192.168.0.245(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, 5 November, 2024 3:12:28 PM
Lease Expires . . . . . : Wednesday, 6 November, 2024 3:12:28 PM
Default Gateway . . . . . : fe80::26e4:ceff:feb6:9ab%5
                             192.168.0.1
DHCP Server . . . . . : 192.168.0.1
DHCPv6 IAID . . . . . : 76331134
DHCPv6 Client DUID. . . . . : 00-01-00-01-29-8F-73-22-E4-A8-DF-AF-9E-8B
DNS Servers . . . . . : 2405:3800:1001:14::160
                             2405:3800:1002:14::176
                             192.168.0.1
                             192.168.0.1
NetBIOS over Tcpip. . . . . : Enabled
```

10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

The query message is a Type “A” query, but it does not have any answer.

```
> Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22
v User Datagram Protocol, Src Port: 3742, Dst Port: 53
    Source Port: 3742
    Destination Port: 53
    Length: 37
    Checksum: 0x5890 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 3]
    [Stream Packet Number: 1]
> [Timestamps]
    UDP payload (29 bytes)
v Domain Name System (query)
    Transaction ID: 0x0003
    > Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
v Queries
    > www.mit.edu: type A, class IN
        [Response In: 20]
```

11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

There is 1 answers provided which is www.mit.edu: type A, class IN, addr 18.7.22.83

```
> Frame 20: 196 bytes on wire (1568 bits), 196 bytes captured (1568 bits)
> Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
> Internet Protocol Version 4, Src: 128.238.29.22, Dst: 128.238.38.160
> User Datagram Protocol, Src Port: 53, Dst Port: 3742
v Domain Name System (response)
    Transaction ID: 0x0003
    > Flags: 0x8580 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 3
    Additional RRs: 3
v Queries
    > www.mit.edu: type A, class IN
v Answers
    v www.mit.edu: type A, class IN, addr 18.7.22.83
        Name: www.mit.edu
        Type: A (1) (Host Address)
        Class: IN (0x0001)
        Time to live: 60 (1 minute)
        Data length: 4
        Address: 18.7.22.83
    > Authoritative nameservers
    > Additional records
        [Request In: 19]
    [Time: 0.016757000 seconds]
```